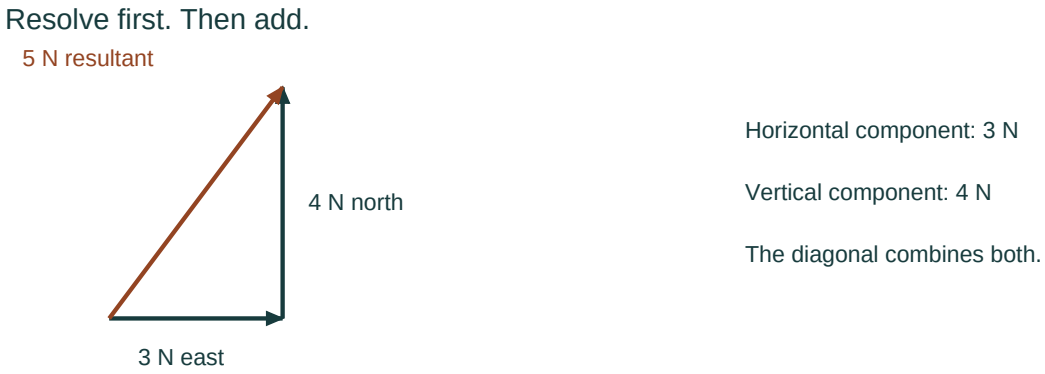
LEVEL 3 / ENGLISH STUDY EDITION

Mechanics and Heat

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.



A clear route from idea to answer

Concept explanations, symbols and conditions, worked steps, recall checks, and two practice tests with explained solutions.

10 lessons | Revised 1 November 2026 | Beginner to exam practice

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Choose units and describe motion
2. Constant acceleration and graphs
3. Resolve vectors before adding
4. Projectiles: one time, two directions
5. Newton laws and friction
6. Circular motion and connected objects
7. Work and kinetic energy
8. Energy conservation and springs
9. Temperature, calorimetry and phase changes
10. Heat transfer and thermal expansion

LESSON 01 / 10

Choose units and describe motion

After this lesson, you can

- Distinguish distance from signed displacement.
- Calculate average speed and average velocity from a route.

Start with: Add and subtract lengths; divide by elapsed time; convert minutes to seconds.

Understand the idea

Position tells you where an object is. Displacement is final position minus initial position; distance counts the entire path. Average velocity uses displacement, while average speed uses distance. A negative velocity gives direction, not a negative speed.

Essential rule

$$\bar{v} = \Delta x / \Delta t$$

$$\text{average speed} = \text{distance} / \Delta t$$

Δx final position minus initial position (m)

Δt elapsed time (s)

$\bar{v}$ average velocity (m/s)

A method that earns marks

1. Choose east or right as positive and sketch the route.
2. Add every path length for distance; subtract start position from finish for displacement.
3. Divide distance and displacement separately by the same total time.
4. Attach a direction to velocity and a unit to every result.

Worked example

A student walks 30 m east and 10 m west in 20 s. Distance is 40 m, displacement is 20 m east, average speed is 2 m/s, and average velocity is 1 m/s east.

1. Take east as positive: the two legs are +30 m and -10 m.
2. Distance counts both legs: 30 + 10 = 40 m; displacement is 30 - 10 = +20 m.
3. Over 20 s, average speed is 40/20 = 2 m/s and average velocity is +20/20 = +1 m/s, east.

Common mistake
Do not replace displacement with distance in a velocity question.

Exam habit: Write the two numerators before dividing: distance for speed, displacement for velocity. This earns method marks even if arithmetic slips.

Check yourself before moving on

Can average speed be nonzero while average velocity is zero?

Answer: Yes. A return trip has zero displacement but a positive total distance.

LESSON 02 / 10

Constant acceleration and graphs

After this lesson, you can

- Select a constant-acceleration equation from the known quantities.
- Interpret slope and area on a velocity-time graph.

Start with: Signed velocity; rectangle and triangle areas; rearranging a simple equation.

Understand the idea

Acceleration measures the change of velocity per second. On a position-time graph, slope is velocity. On a velocity-time graph, slope is acceleration and signed area is displacement. The constant-acceleration equations require acceleration to remain constant.

Essential rule

$$v = u + at$$
$$s = ut + \tfrac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

u initial velocity (m/s)

v final velocity (m/s)

a constant acceleration (m/s^2)

s displacement (m)

t elapsed time (s)

When this applies

The equations in this lesson assume constant acceleration throughout the stated interval.

A method that earns marks

1. List the known quantities with signs and identify the unknown.
2. Check that the acceleration is constant before using the equations.
3. Choose an equation without an extra unknown, then substitute with units.
4. Use graph slope or area as an independent interpretation of the result.

Worked example

A car starts from rest and accelerates at 3 m/s^2 for 4 s. Its speed is 12 m/s and displacement is 24 m. The triangular area under its velocity-time graph gives the same displacement.

1. Starts from rest means u = 0; the example gives a = 3 m/s^2 and t = 4 s.
2. The velocity gain is at = 12 m/s, so v = 12 m/s.
3. The displacement is one half times 3 times 4^2 = 24 m.
4. On the velocity-time graph the triangle has base 4 s and height 12 m/s; its area is also 24 m.

Common mistake
A straight horizontal velocity-time graph means zero acceleration, not zero velocity.

Exam habit: For a graph, state which axes you used before calculating slope or area; their units identify acceleration or displacement.

Check yourself before moving on

A car has velocity +8 m/s and acceleration -2 m/s^2. Is it initially speeding up?

Answer: No. Acceleration opposes its current velocity, so its speed initially decreases.

LESSON 03 / 10

Resolve vectors before adding

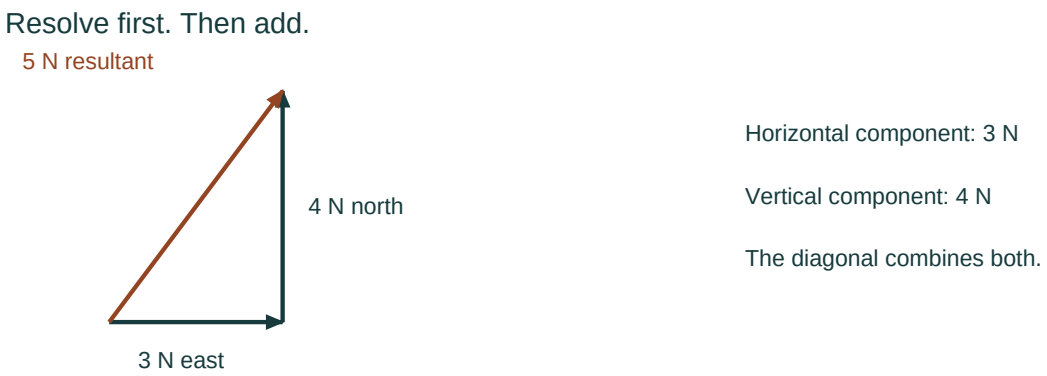
After this lesson, you can

- Add perpendicular vectors using signed components.
- Give a resultant direction in the correct quadrant.

Start with: Right-triangle trigonometry; Pythagoras; compass directions.

Understand the idea

A vector has size and direction. Two vectors must be added through their signed components, not by simply adding magnitudes. The resultant points from the initial tail to the final head in a head-to-tail drawing.



Perpendicular forces add as vectors. A 3 N east component and 4 N north component give a 5 N resultant; the balancing force points the other way.

Essential rule

$$A_x = A\cos\theta$$

$$A_y = A\sin\theta$$

$$R = \sqrt{R_x^2 + R_y^2}$$

A_x horizontal vector component (same as vector)

A_y vertical vector component (same as vector)

R resultant magnitude (same as vector)

θ angle from the stated axis (degrees or radians)

A method that earns marks

1. Draw axes and identify the reference axis for each angle.
2. Resolve each vector and assign signs to its components.
3. Add horizontal components and vertical components separately.
4. Calculate magnitude and use component signs to choose the direction.

Worked example

A 3 N force east and a 4 N force north produce a 5 N resultant, 53.1 degrees north of east. The balancing force is 5 N in the opposite direction.

1. East and north are perpendicular, so the example has Rx = 3 N and Ry = 4 N.
2. The magnitude is the square root of $3^2 + 4^2$, giving 5 N.
3. The angle satisfies $\tan\theta = 4/3$, so $\theta = 53.1$ř north of east.
4. A balancing force reverses both components: 3 N west and 4 N south, with the same 5 N magnitude.

Common mistake
The inverse tangent alone can return the wrong quadrant.

Exam habit: Report both an angle and its reference direction; an angle alone can leave the quadrant ambiguous.

Check yourself before moving on

Does a vector with negative horizontal and positive vertical components point northeast?

Answer: No. It points northwest, in quadrant II.

LESSON 04 / 10

Projectiles: one time, two directions

After this lesson, you can

- Separate a projectile into horizontal and vertical motions.
- Use the shared flight time to calculate horizontal range.

Start with: Constant-acceleration motion; squaring and square roots; horizontal and vertical directions.

Understand the idea

Ignoring air resistance, horizontal acceleration is zero and vertical acceleration is downward. Horizontal and vertical motion share the same time. At the top, only the vertical velocity becomes zero; horizontal velocity continues.

Essential rule

$$v_x = u\cos\theta$$

$$v_y = u\sin\theta - gt$$

$$y = u\sin\theta\, t - \tfrac{1}{2}gt^2$$

u_x initial horizontal velocity (m/s)

u_y initial vertical velocity (m/s)

g gravitational acceleration magnitude (m/s^2)

t flight time (s)

When this applies

Ignore air resistance and take gravity as uniform; launch and landing heights need not match.

A method that earns marks

1. Draw the launch point and select a positive vertical direction.
2. Write horizontal motion with zero acceleration and vertical motion with gravity.
3. Use the height information to find the positive physical flight time.
4. Use this same time in the horizontal displacement equation.

Worked example

A ball leaves a horizontal table at 3 m/s and takes 0.4 s to land. Range = 1.2 m and drop = 0.5 times 9.8 times 0.4 squared = 0.784 m.

1. A horizontal launch has initial vertical velocity zero; its horizontal velocity remains 3 m/s.
2. The stated time is 0.4 s, so horizontal range is 3 times 0.4 = 1.2 m.
3. The vertical drop is one half times 9.8 times 0.4^2 = 0.784 m.
4. The unequal range and drop are consistent because the two directions obey different motion equations.

Common mistake
Do not set the total velocity to zero at the highest point.

Exam habit: If you choose upward positive, the downward displacement and acceleration must both carry negative signs.

Check yourself before moving on

At the highest point of an angled projectile, which velocity component is zero?

Answer: Only the vertical component; horizontal velocity remains constant in this model.

LESSON 05 / 10

Newton laws and friction

After this lesson, you can

- Construct a free-body diagram for one selected object.
- Use net force and friction to find acceleration.

Start with: Signed addition; mass and weight; perpendicular directions.

Understand the idea

A free-body diagram shows only forces acting on the selected object. Net force causes acceleration. Action-reaction forces act on different objects and therefore do not cancel on one free-body diagram. Static friction adjusts up to its maximum; kinetic friction applies during sliding.

Essential rule

$$\sum F = ma$$
$$W = mg$$
$$f_k = \mu_k N$$
$$f_s \leq \mu_s N$$

m object mass (kg)

N normal contact force (N)

μ_k coefficient of kinetic friction (dimensionless)

a acceleration (m/s^2)

When this applies

The example treats the block as sliding and the given 4 N as kinetic friction.

A method that earns marks

1. Isolate one object and draw only the forces acting on it.
2. Choose axes and use the perpendicular force balance to obtain the normal force if needed.
3. Choose static or kinetic friction from the stated motion.
4. Add signed forces along the motion and divide the net force by mass.

Worked example

A 2 kg block on a horizontal surface is pulled with 10 N. With kinetic friction 4 N, the net force is 6 N and acceleration is 3 m/s^2.

1. Choose the 10 N pull as positive; the 4 N sliding friction acts in the opposite direction.
2. The horizontal net force is 10 - 4 = 6 N.
3. For the 2 kg block, a = 6/2 = 3 m/s^2 in the direction of the pull.
4. Weight and normal force balance vertically under the horizontal-surface assumptions, so they do not enter the horizontal sum.

Common mistake
The normal force is not always mg: angled pulls and inclines change it.

Exam habit: Show the signed force equation before substitution; do not write the applied force alone as ma.

Check yourself before moving on

A sliding block has a 7 N pull and 7 N friction. Must it be at rest?

Answer: No. Zero net force gives zero acceleration; it may continue moving at constant velocity.

LESSON 06 / 10

Circular motion and connected objects

After this lesson, you can

- Identify the real forces that supply inward acceleration.
- Set up separate force equations for connected masses.

Start with: Newton second law; force diagrams; uniform circular motion directions.

Understand the idea

Changing direction is acceleration even at constant speed. Centripetal means inward; it names the required net force, not an extra physical force. In an ideal taut string over a massless frictionless pulley, connected objects share acceleration magnitude.

Essential rule

$$a_r = v^2/r$$

$$\sum F_{\text{inward}} = mv^2/r$$

v speed along the circle (m/s)

r circle radius (m)

a_r inward acceleration (m/s^2)

T string tension (N)

When this applies

Connected-object formulas assume a taut massless string and an ideal frictionless pulley when stated.

A method that earns marks

1. For circular motion mark the centre and select inward as a positive radial direction.
2. Sum the actual radial force components and equate them to mv^2/r.
3. For connected objects draw a diagram for each mass and use compatible acceleration directions.
4. Use the ideal string constraint to share acceleration magnitude, then solve for acceleration and tension.

Worked example

A 2 kg object moves at 3 m/s in a circle of radius 1.5 m. Inward acceleration = 6 m/s^2 and inward net force = 12 N.

1. In the circular example, v = 3 m/s and r = 1.5 m.
2. Squaring speed first gives ar = 9/1.5 = 6 m/s^2, directed toward the centre.
3. Multiplying by the 2 kg mass gives 12 N inward net force.
4. The problem would need a physical setup to decide whether tension, gravity or another force provides those 12 N.

Common mistake
Do not add a separate centripetal force to tension or gravity.

Exam habit: Do not put tension and an extra centripetal force on the same diagram; centripetal describes the inward net force.

Check yourself before moving on

If the speed remains fixed and radius doubles, what happens to inward acceleration?

Answer: It halves because v^2/r is inversely proportional to radius at fixed speed.

LESSON 07 / 10

Work and kinetic energy

After this lesson, you can

- Determine the sign of work from force and displacement directions.
- Calculate net work from a change in kinetic energy.

Start with: Cosine of common angles; squaring speed; positive and negative energy transfers.

Understand the idea

Work transfers energy when a force has a component along displacement. A perpendicular force does no work. Net work equals the change in kinetic energy. For a varying force along a line, work is the signed area under the force-position graph.

Essential rule

$$W = Fd\cos\theta$$

$$K = \tfrac{1}{2}mv^2$$

$$W_{\text{net}} = K_f - K_i$$

W work transferred by a force (J)

F force magnitude (N)

d displacement magnitude (m)

K kinetic energy (J)

A method that earns marks

1. Identify the displacement and the angle between it and the force.
2. Use only the force component along displacement to calculate work.
3. Add work from all forces for net work, or use the kinetic-energy change.
4. Check that positive net work corresponds to an increase in speed.

Worked example

A 10 N force pulls through 3 m at 60 degrees to displacement. Work = 10 times 3 times 0.5 = 15 J.

1. The example states an angle of 60ř between the 10 N force and 3 m displacement.
2. Only 10 cos 60ř = 5 N acts along the displacement.
3. Work by this force is 5 times 3 = 15 J.
4. The remaining perpendicular component contributes zero work; 15 J is not automatically net work if other forces also do work.

Common mistake
Use the force-displacement angle, not an unrelated angle in the drawing.

Exam habit: When using the energy theorem, square each speed before subtracting; $(vf - vi)^2$ is not $vf^2 - vi^2$.

Check yourself before moving on

Can a nonzero force do zero work during a displacement?

Answer: Yes. A force perpendicular to the displacement does zero work.

LESSON 08 / 10

Energy conservation and springs

After this lesson, you can

- Track kinetic, gravitational and spring energy between two states.
- Identify when a loss term is required in an energy balance.

Start with: Work and kinetic energy; choosing a height reference; spring compression or extension.

Understand the idea

Energy can move between kinetic, gravitational and elastic stores. Mechanical energy stays constant only when the model excludes losses or other transfers. Choose a height reference; only height changes affect the result.

Essential rule

$$K_i + U_i + W_{\text{other}} = K_f + U_f$$

$$U_g = mgh$$

$$U_s = \tfrac{1}{2}kx^2$$

U_g gravitational potential energy (J)

U_s elastic potential energy (J)

k spring stiffness (N/m)

x spring deformation from natural length (m)

When this applies

The falling-mass example neglects air resistance and other energy transfers; spring-energy formulas require the proportional linear range.

A method that earns marks

1. Name initial and final states and choose a zero-height level.
2. Write the energy balance before removing terms that are zero.
3. Include friction or other transfers if the model contains them.
4. Solve for the unknown and check the physical sign and units.

Worked example

A mass drops 5 m from rest without losses. With g = 10 m/s^2, its final speed follows v squared = 2gh = 100, giving 10 m/s independent of mass.

1. Set the final height to zero; the initial height is then 5 m.
2. The object starts from rest, so its initial energy is gravitational: mg times 5.
3. With no losses, mg times 5 equals one half mv^2; dividing by mass shows why mass cancels.
4. Using g = 10 m/s^2 gives v^2 = 100 and speed v = 10 m/s.

Common mistake
Spring extension is measured from natural length, not total length.

Exam habit: For spring energy convert centimetres to metres before squaring the deformation.

Check yourself before moving on

Would the same loss-free 5 m fall give a heavier object a larger final speed?

Answer: No. Mass cancels from mgh = one half mv^2.

LESSON 09 / 10

Temperature, calorimetry and phase changes

After this lesson, you can

- Separate temperature change from phase change in a heating problem.
- Apply an energy balance with consistent mass and temperature units.

Start with: Energy in joules; mass conversion; subtraction of temperatures.

Understand the idea

Temperature describes thermal state; heat is energy transferred because of a temperature difference. During a phase change of a pure substance at fixed pressure, energy can transfer without a temperature change. In an insulated mixing model, heat lost equals heat gained.

Essential rule

$$Q = mc\Delta T$$

$$Q = mL$$

$$T_K = T_C + 273.15$$

Q energy transferred as heat (J)

c specific heat capacity (J/(kg K))

L specific latent heat (J/kg)

ΔT temperature change (K or řC difference)

A method that earns marks

1. Identify the initial and final phase and temperature.
2. Split the process into warming, phase-change and further-warming stages where needed.
3. Use mcΔT for each warming stage and mL for each phase change.
4. Add the energy requirements or balance heat gained and lost in an insulated mixture.

Worked example

Heating 0.20 kg of water by 10 degrees C needs Q = 0.20 times 4200 times 10 = 8400 J. Melting ice requires a separate latent-heat term.

1. The water remains liquid while warming, so the relevant relation is Q = mcΔT.
2. A rise of 10řC is a temperature difference of 10 K.
3. For 0.20 kg and c = 4200 J/(kg K), Q = 0.20 times 4200 times 10 = 8400 J.

4. If the process also melts ice, that extra stage needs mL; it is not included in the 8400 J.

Common mistake
A temperature difference of 10 degrees C equals 10 K; an absolute temperature does not.

Exam habit: On a heating curve label sloping segments mcΔT and horizontal phase-change segments mL.

Check yourself before moving on

Does liquid water warming from 20řC to 30řC have $\Delta T = 283.15$ K?

Answer: No. $\Delta T = 10$ K. The Celsius-to-kelvin offset cancels when subtracting two temperatures.

LESSON 10 / 10

Heat transfer and thermal expansion

After this lesson, you can

- Classify heat transfer as conduction, convection or radiation.
- Predict rate changes and calculate small linear expansion.

Start with: Direct and inverse proportionality; temperature differences; metres and millimetres.

Understand the idea

Conduction transfers energy through matter without bulk transport; convection uses fluid motion; radiation uses electromagnetic waves and can cross vacuum. Most solids expand on warming. A thermal insulator reduces transfer but does not create cold.

Essential rule

$$\Delta L = \alpha L_0 \Delta T$$

$$P_{\text{cond}} = kA\Delta T/L$$

$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_s^4)$$

P heat-transfer rate (W)

k thermal conductivity (W/(m K))

A transfer surface area (m^2)

L wall thickness (m)

α linear expansion coefficient (K^{-}ź)

When this applies

The wall comparison uses steady one-dimensional conduction with unchanged material and temperature difference.

A method that earns marks

1. Identify the physical mechanism and select the matching relation.
2. For a comparison, hold every unstated quantity fixed and change only the named factor.
3. For expansion use the original length and the temperature rise.
4. Use kelvin for absolute temperatures in radiation and convert the final length if requested.

Worked example

Doubling the thickness of an otherwise unchanged wall halves its conductive heat-transfer rate. Doubling the area doubles that rate.

1. For conduction, the same wall material and temperature difference give P proportional to A/L.
2. Doubling thickness replaces L by 2L, so P becomes P/2.
3. Doubling area instead gives 2P.
4. If both area and thickness double together, the factors cancel and the rate is unchanged.

Common mistake
Use kelvin in fourth-power radiation equations; Celsius produces incorrect results.

Exam habit: For a ratio question explicitly state what stays constant; otherwise the proportionality is incomplete.

Check yourself before moving on

A wall is made three times thicker with all other quantities unchanged. What is its new conduction rate?

Answer: One third of the original rate.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Mechanics and Heat / Test A

25 marks | Suggested time: 25 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. A runner completes one lap and returns to the start. Average velocity is:

   A. Always positive

   B. Impossible to define

   C. Zero

   D. Equal to average speed

2. The area under a velocity-time graph gives:

   A. Displacement

   B. Force

   C. Speed at the final instant

   D. Acceleration

3. Two perpendicular forces of 6 N and 8 N have resultant magnitude:

   A. 10 N

   B. 48 N

   C. 2 N

   D. 14 N

4. At the top of a projectile path, acceleration is:

   A. Zero

   B. Horizontal

   C. Upward

   D. Downward with magnitude g

5. An object moving at constant velocity has:

   A. A forward net force

   B. Increasing acceleration

   C. Zero net force

   D. No forces acting at all

Part II: Show your reasoning

1. A cart travels 12 m right then 4 m left in 4 s. Find distance, displacement, average speed and average velocity.

2. A car slows from 20 to 8 m/s in 4 s with constant acceleration. Find acceleration and displacement.

3. Find the magnitude and direction of a vector with components -3 m and +4 m.

4. A ball is projected horizontally at 5 m/s from a height of 4.9 m. Ignore air resistance and use $g = 9.8\ m/s^2$. Find flight time and horizontal range.

5. A 5 kg block slides to the right on a horizontal floor with coefficient of kinetic friction 0.2. It is pulled horizontally to the right with 20 N. Use $g = 10\ m/s^2$. Find normal force, friction and acceleration.

Mechanics and Heat / Test B

25 marks | Suggested time: 25 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Doubling speed at fixed radius changes centripetal acceleration by a factor of:

   A. Two

   B. Four

   C. Eight

   D. One half

2. A force perpendicular to displacement does work equal to:

   A. Fd

   B. Zero

   C. -Fd

   D. Half of Fd

3. If the extension of a Hooke-law spring doubles while k stays constant, its stored energy becomes:

   A. Four times as large

   B. Twice as large

   C. Unchanged

   D. Half as large

4. During melting at fixed pressure, added heat mainly:

   A. Raises temperature continuously

   B. Changes phase

   C. Decreases mass

   D. Disappears

5. Which heat-transfer mechanism can operate through a vacuum?

   A. Fluid mixing

   B. Convection

   C. Conduction through air

   D. Radiation

Part II: Show your reasoning

1. An ideal Atwood machine has masses 2 kg and 3 kg. Use g = 10 m/s^2. Find acceleration and tension.

2. A 2 kg object speeds up from 3 to 5 m/s. Find the net work done.

3. A spring of constant 200 N/m is compressed by 0.10 m and releases a 0.50 kg block on a frictionless horizontal surface. Find its speed when the spring returns to natural length.

4. How much energy heats 0.50 kg of water from 20 to 30 degrees C? Use c = 4200 J/(kg K).

5. A rod is 2 m long and has expansion coefficient 0.000012 per K. It warms by 50 K. Find the increase in length in metres and millimetres.

Mechanics and Heat / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** The net displacement is zero even though the distance travelled is not.
2. **A.** Area has units (m/s) times s = m and includes the sign of velocity.
3. **A.** The Pythagorean calculation gives the square root of 36+64 = 10 N.
4. **D.** Gravity continues to act at the top.
5. **C.** Constant velocity means zero acceleration, so forces balance.

Part II: Written solutions

1. Distance = 16 m; displacement = +8 m; speed = 16/4 = 4 m/s; velocity = 8/4 = +2 m/s. Award one mark for each quantity.
2. a = (8-20)/4 = -3 m/s^2. s = [(20+8)/2] times 4 = 56 m. Two marks for each calculation, including the sign and unit.
3. Magnitude = 5 m. The vector is in quadrant II: 126.9 degrees counterclockwise from positive x, or 53.1 degrees north of west. Two marks for magnitude and two for a correct direction.
4. 4.9 = 0.5 times 9.8 times t^2, so t = 1 s. Range = 5 times 1 = 5 m. Two marks per result with the equation shown.
5. N = 50 N, friction = 0.2 times 50 = 10 N, net force = 10 N, a = $1\frac{0}{5}$ = 2 m/s^2. One mark for each step.

Mechanics and Heat / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** The speed is squared.
2. **B.** The cosine of 90 degrees is zero.
3. **A.** Spring energy is proportional to extension squared.
4. **B.** Energy changes the phase while temperature remains at the melting point.
5. **D.** Electromagnetic radiation needs no material medium.

Part II: Written solutions

1. Net driving force on the two-mass system is (3-2) times 10 = 10 N. Total mass = 5 kg, so a = 2 m/s^2. For the rising 2 kg mass, T-20 = 2 times 2, hence T = 24 N. Two marks per result.
2. Net work = 0.5 times 2 times (25-9) = 16 J. Give marks for the energy theorem, both squared speeds, substitution and unit.
3. Spring energy = 0.5 times 200 times 0.10 squared = 1 J. Set 1 = 0.5 times 0.50 times v squared to get v = 2 m/s. Two marks for each stage.
4. Temperature rise = 10 K. Q = mc times temperature rise = 0.50 times 4200 times 10 = 21000 J. Four marks: model, temperature difference, substitution, result/unit.
5. Increase = 0.000012 times 2 times 50 = 0.0012 m = 1.2 mm. Award marks for formula, substitution, metres and conversion.

Source and coverage notes

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

Source folder: https://drive.google.com/drive/folders/1-FxWftel94cEQ9LPiUPpzLy1OYMKhGVg
The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.